

Superstore Sales & Profit Analysis

An end-to-end exploratory analysis of a US retail superstore's order data — uncovering monthly sales trends, category and sub-category performance, profit drivers, customer-segment behavior, and profit efficiency, built entirely with pandas, NumPy, and Matplotlib.

SUMMARY

The Sample Superstore dataset contains order line items from a US-based retail store, spanning three product categories (Furniture, Office Supplies, Technology) and 17 sub-categories, sold across three customer segments (Consumer, Corporate, Home Office). This case study analyzes monthly sales and profit trends, category and sub-category performance, customer-segment behavior, and profit efficiency using pandas, NumPy, and Matplotlib.

OBJECTIVE

- Identify monthly sales & profit trends and peak/trough months
- Compare performance across categories and sub-categories
- Understand profitability by customer segment
- Measure how efficiently sales convert into profit

TOOLS USED

Python	pandas	NumPy
Matplotlib	Jupyter Notebook	

Case Study Questions

Seven questions this analysis set out to answer.

- 01** Calculate the monthly sales of the store and identify which month had the highest sales and which month had the lowest sales.

- 02** Analyze sales based on product categories and determine which category has the lowest sales and which category has the highest sales.

- 03** The sales analysis needs to be done based on sub-categories.

- 04** Analyze the monthly profit from sales and determine which month had the highest profit.

- 05** Analyze the profit by category and sub-category.

- 06** Analyze the sales and profit by customer segment.

- 07** Analyze the sales-to-profit ratio.